

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

- First Line Contains Integer n – Size of array
- Next n lines Contains n numbers – Elements of an array
- Last Line Contains Integer x – Sum Value

Output Format

- First Line Contains Integer – Element1
- Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2 void findPair(int arr[], int l, int r, int x) {
3     if (l >= r) { printf("No\n"); return; }
4     int sum = arr[l] + arr[r];
5     if (sum == x) { printf("%d\n%d\n", arr[l], arr[r]); return; }
6     if (sum < x) findPair(arr, l + 1, r, x);
7     else findPair(arr, l, r - 1, x);
8 }
9 int main() {
10     int n, x;
11     if (scanf("%d", &n) != 1) return 0;
12     int arr[n];
13     for (int i = 0; i < n; i++) scanf("%d", &arr[i]);
14     if (scanf("%d", &x) != 1) return 0;
15     findPair(arr, 0, n - 1, x);
16     return 0;
17 }
```

	Input	Expected	Got	
✓	4	4	4	✓
	2	10	10	
	4			
	8			
	10			
	14			
✓	5	No	No	✓
	2			
	4			
	6			
	8			
	10			
	100			

Passed all tests! ✓

Correct